

EXERCISE

GIVEN THE FOLLOWING SYSTEM:

$$\begin{aligned} x(t+1) &= \frac{2}{5}x(t) + n_1(t) \\ y(t) &= 3x(t) + n_2(t) \end{aligned}$$

$$\begin{aligned} n_1 &\sim \text{WN}(0, \frac{123}{125}) \\ n_2 &\sim \text{WN}(0, 1) \end{aligned} \quad n_1 \perp n_2$$

⇒ WRITE THE EQUATIONS OF THE 1-STEP AHEAD KALMAN PREDICTOR

THE EQUATIONS (SEE THEORY) ARE:

$$\hat{x}(t+1|t) = F \cdot \hat{x}(t|t-1) + k(t) \cdot e(t) \quad \text{STATE PREDICTION}$$

$$\hat{y}(t|t-1) = H \cdot \hat{x}(t|t-1) \quad \text{OUTPUT PREDICTION}$$

$$e(t) = y(t) - \hat{y}(t|t-1) \quad \text{INNOVATION}$$

$$k(t) = (F \cdot P(t) \cdot H^T + V_{12}) (H \cdot P(t) \cdot H^T + V_2)^{-1} \quad \text{KALMAN GAIN}$$

$$P(t+1) = (F \cdot P(t) \cdot F^T + V_1) - (F \cdot P(t) \cdot H^T + V_{12}) (H \cdot P(t) \cdot H^T + V_2)^{-1} (F \cdot P(t) \cdot H^T + V_{12})^T \quad \text{DRS}$$

LET'S EXPRESS THE "MISSING" DATA:

$$F = \frac{2}{5}, \quad H = 3, \quad V_1 = \frac{123}{125}, \quad V_2 = 1, \quad V_{12} = 0 \quad (\text{SINCE } n_1 \perp n_2)$$

PLUGGING THE VALUES INSIDE THE PREVIOUS GENERAL EQUATIONS WE HAVE:

$$\hat{x}(t+1|t) = \frac{2}{5} \hat{x}(t|t-1) + k(t) \cdot e(t)$$

$$\hat{y}(t|t-1) = 3 \hat{x}(t|t-1)$$

$$e(t) = y(t) - \hat{y}(t|t-1)$$

$$k(t) = \frac{\frac{6}{5} \cdot P(t)}{9P(t) + 1}$$

$$P(t+1) = \frac{1127 \cdot P(t) + 123}{125(9 \cdot P(t) + 1)}$$

⇒ WRITE THE EQUATIONS OF THE STEADY-STATE 1-STEP AHEAD PREDICTOR

STEP # 1 - COMPUTE (IF EXISTS) FOR THE ARG: $P(t) = P(t+1) = \bar{P}$

$$\text{ARG: } \bar{P} = \frac{1127 \cdot \bar{P} + 123}{125 \cdot 9 \cdot \bar{P} + 125}$$

⇓

$$1125 \bar{P}^2 + 125 \bar{P} - 1127 \bar{P} - 123 = 0$$

$$1125 \bar{P}^2 - 1002 \bar{P} - 123 = 0$$

$$\bar{P}_{1,2} = \frac{501 \pm 624}{1125}$$

$$\bar{P}_1 = 1$$

$$\bar{P}_2 = \frac{-123}{1125}$$

Thus: $\bar{K} = \left(\frac{2}{5} \cdot 1 \cdot 3 \right) \left(3 \cdot 1 \cdot 3 + 1 \right)^{-1} = \frac{3}{25}$

$\uparrow$
 $P(t) = \bar{P}$

THIS MEANS THAT OUR STEADY-STATE 1-STEP AHEAD KALMAN PREDICTOR IS:

$$\hat{x}(t+1|t) = \left[\frac{2}{5} - \frac{9}{25} \right] \hat{x}(t|t-1) + \frac{3}{25} y(t)$$

$$\hat{y}(t+1|t) = 3 \cdot \hat{x}(t+1|t)$$

STEP #2 - CONVERGENCE OF DRS TO ARS

(A) THEOREMS

TH #1

- $V_{12} = 0$ ✓
 - F IS A.S. ✓
- $\Rightarrow$
- ARS HAS UNIQUE SOLUTION $\bar{P} \geq 0$
 - $P(t) \rightarrow \bar{P} \quad \forall P(0) = P_0$
 - THE PREDICTOR IS A.S.

TH #2

- $V_{12} = 0$ ✓
 - (F, H) IS OBSERVABLE ✓
 - (F, P) IS CONTROLLABLE ✓
- $\Rightarrow$
- ARS HAS UNIQUE SOLUTION $\bar{P} > 0$
 - $P(t) \rightarrow \bar{P} \quad \forall P(0) = P_0$
 - THE PREDICTOR IS A.S.

↑
NOISE MATRIX
(STATE EQUATIONS)

$$x(t+1) = Fx(t) + \underbrace{P \tilde{N}_1(t)}_{N_1(t)}$$

↑
IDENTITY MATRIX

$$\tilde{N} \sim WN(0, I)$$

IN OUR CASE $P = \sqrt{\frac{123}{125}}$

SINCE

$$x(t+1) = \dots + \tilde{N}_1 \quad \tilde{N}_1 \sim WN(0, \frac{123}{125})$$

$\Updownarrow$

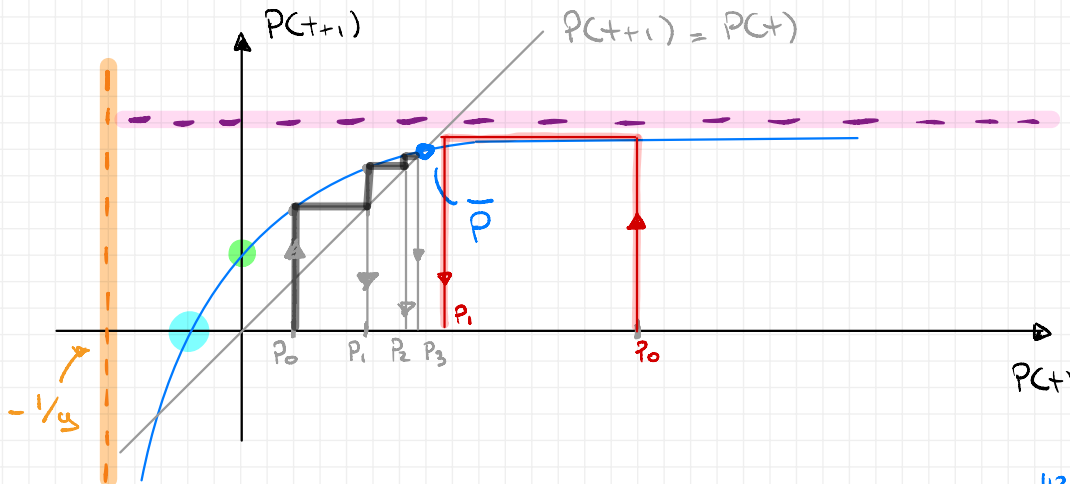
$$x(t+1) = \dots + \sqrt{\frac{123}{125}} \cdot \tilde{N}_1 \quad \tilde{N}_1 \sim WN(0, 1)$$

- $O = [H] \neq 0 \Rightarrow$ SYSTEM IS OBSERVABLE
- $R = [P] \neq 0 \Rightarrow$ SYSTEM IS CONTROLLABLE

② OTHER TOOLS FOR STUDYING THE CONVERGENCE OF DRG $\rightarrow$ ARS

1. "LINEAR STABILITY ANALYSIS": SOMETIMES THE DRG IS A LINEAR DYNAMIC EQUATION

2. GRAPHICAL METHOD FOR NONLINEAR AND SCALAR DRG



ON THIS DIAGRAM WE REPRESENT THE DRG: $P(t+1) = \frac{1127 P(t) + 123}{125 (2 \cdot P(t) + 1)}$ DRG

LET'S REPRESENT ON THE DIAGRAM THE DRG:

• $P(t) = -1/9$ VERTICAL ASYMPTOTE

• $P(t) \rightarrow +\infty \quad P(t+1) \rightarrow \frac{1127}{125 \cdot 2} = \frac{1127}{1125} > 1$ HORIZONTAL ASYMPTOTE

• $P(t) = 0 \rightarrow P(t+1) = \frac{123}{125}$

• $P(t+1) = 0 \rightarrow P(t) = -\frac{123}{1127} \sim -0.1$

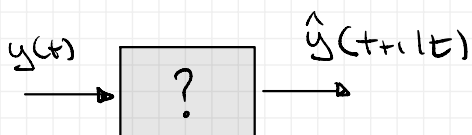
THE GRAPHICAL ANALYSIS* OF DRG TELLS US THAT (AS WE KNOW) DRG CONVERGES TO THE SOLUTION OF THE ARS.

* GRAPHICAL ANALYSIS: SIMULATION OF THE EVOLUTION OF $P(t)$, USING THE DIAGRAM TO COMPUTE $P(t+1)$, $P(t+2)$, ...

$$P(t) \rightarrow \bar{P} \quad \forall P_0 \geq 0$$

$\Rightarrow$ COMPUTES THE T.F. FROM THE MEASURED OUTPUT TO THE 1-STOP AROUND STEADY-STATE PREDICTED OUTPUT AND COMPARES IT WITH THE RESULTS OF K.W. PREDICTION

WE ARE LOOKING FOR:



LET'S START WITH THE KALMAN PREDICTOR:

$$\left. \begin{aligned} \hat{x}(t+1|t) &= 1/25 \hat{x}(t|t-1) + 3/25 \cdot y(t) \\ \hat{y}(t|t-1) &= 3 \cdot \hat{x}(t|t-1) \end{aligned} \right\} \text{KALMAN PREDICTOR}$$

WE SIMPLY REWELL THAT:

$$\hat{y}(t|t-1) = 3 \cdot \hat{x}(t|t-1) \rightarrow \hat{y}(t+1|t) = 3 \cdot \hat{x}(t+1|t)$$

SO, USING THIS STATE EQUATION WE HAVE:

$$\hat{x}(t+1|t) = 1/25 \cdot \underbrace{z^{-1} \cdot \hat{x}(t+1|t)}_{\hat{x}(t|t-1)} + 3/25 y(t)$$

$\Downarrow$

$$\hat{x}(t+1|t) = \frac{3/25}{1 - 1/25 \cdot z^{-1}} \cdot y(t)$$

$\Downarrow$

$$\hat{y}(t+1|t) = \frac{9/25}{1 - 1/25 \cdot z^{-1}} \cdot y(t)$$

NOW WE WANT TO COMPUTE THE K.W. PREDICTOR. WE HAVE TO EXPRESS $y(t)$ AS:

$$y(t) \rightarrow \boxed{?} \rightarrow y(t) \quad y(t) \sim WN(0, \lambda^2)$$

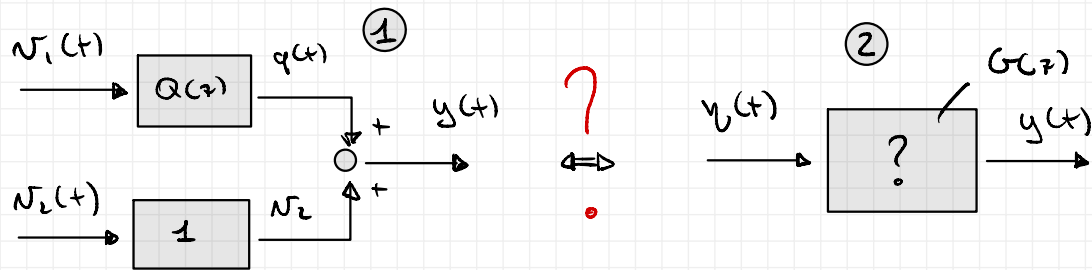
WE NEED TO CONVERT THE STATE-SPACE DESCRIPTION OF THE SYSTEM INTO AN I/O T.F. ONE.

LET'S START BY COMPUTING THE T.F. OF THE ORIGINAL SYSTEM

$$\begin{aligned} x(t+1) &= 2/5 x(t) + v_1(t) \\ y(t) &= 3 x(t) + v_2(t) \end{aligned} \xrightarrow{z} \begin{aligned} x(t) \cdot z &= 2/5 x(t) + v_1(t) \\ y(t) &= 3 \cdot x(t) + v_2(t) \end{aligned}$$

$$x(t) = \frac{1}{z - 2/5} \cdot v_1(t) \rightarrow y(t) = \frac{3}{z - 2/5} \cdot v_1(t) + v_2(t) \quad (Q(z))$$

WE GOT A DESCRIPTION OF $y(t)$ IN T.F. FORM, BUT WITH 2-NOISES!



WE COMPUTE $G(z)$ BY MAKING THE PSD OF THE TWO PROCESSES EQUAL

$$P_y = P_q + P_{N_2} + 0 \quad \sim N_1(t) \text{ and } N_2(t) \text{ are } \perp$$

$$= |Q(e^{j\omega})|^2 \cdot P_{N_1} + P_{N_2}$$

$$= \frac{3 \cdot 3}{(e^{j\omega} - 2/5)(e^{-j\omega} - 2/5)} \cdot \frac{123}{125} + 1$$

$$P_y = \frac{9 \cdot 123 / 125 + 29 / 25 - 4/5 \cos \omega}{29 / 25 - 4/5 \cos \omega}$$

IF WE ASSUME $G(z)$ BEING A GENERIC ARMA (1,1) PROCESS WE GOT:

$$G(z) = \frac{1 + cz^{-1}}{1 + az^{-1}} \xrightarrow{P} P = \frac{(1+c^2) + 2 \cdot c \cdot \cos \omega}{(1+a^2) + 2 \cdot a \cdot \cos \omega} \cdot \lambda^2$$

$$\text{DEN: } \begin{cases} (1+a^2) = 29/25 \\ 2a = -4/5 \end{cases} \Rightarrow a = -2/5$$

$$\text{NUM: } \begin{cases} (1+c^2) \lambda^2 = \frac{9 \cdot 123}{125} + \frac{29}{25} \\ 2 \cdot c \cdot \lambda^2 = -4/5 \end{cases} \Rightarrow c_{1,2} = \begin{cases} -25 \Rightarrow \lambda^2 = 2/125 \\ -1/25 \Rightarrow \lambda^2 = 10 \end{cases}$$

SO WE HAVE TWO POSSIBLE DESCRIPTIONS:

$$G_1 = \frac{1 - 25z^{-1}}{1 - 2/5 z^{-1}} \cdot y(t)$$

$$y(t) \sim \text{WN}(0, 2/125)$$

$$G_2 = \frac{1 - 1/25 z^{-1}}{1 - 2/5 z^{-1}} \cdot y(t)$$

$$y(t) \sim \text{WN}(0, 10)$$

THIS IS NOT
IN CANONICAL FORM

THIS IS IN CANONICAL FORM!

LET'S COMPUTE THE 1-STEP AHEAD PREDICTOR:

$$\hat{y}(t+1|t) = \frac{C(z) - A(z)}{C(z)} y(t+1)$$

$$= \frac{2/5 - 1/25}{1 - 1/25 z^{-1}} y(t)$$

$$\hat{y}(t+1|t) = \frac{9/25}{1 - 1/25 z^{-1}} y(t)$$

SAME RESULTS AS KALMAN!

⇒ COMPUTES THE STEADY-STATE OUTPUT 1-STEP AHEAD PREDICTION ERROR VARIANCES

$$P(t) = \text{Var} [x(t) - \hat{x}(t|t-1)]$$

$$\text{Var} [y(t) - \hat{y}(t|t-1)]$$

INTERPRETATION OF $P(t)$ FROM KALMAN THEORY

QUANTITY WE ARE INTERESTED IN

WE NEED TO EXPRESS $\text{Var} [E_y(t+1)]$ AS A FUNCTION OF $\text{Var} [E_x(t+1)]$

$$\text{Var} [E_y(t+1)] = \mathbb{E} [(y(t+1) - \hat{y}(t+1|t))^2]$$

$$= \mathbb{E} [(\underbrace{3 \cdot x(t+1) + n_2(t+1)}_{\text{OUTPUT EQ. SYSTEM}} - \underbrace{3 \cdot \hat{x}(t+1|t)}_{\hat{y}(t+1|t) \text{ OUTPUT EQUATION PREDICTOR}})^2]$$

$$= \mathbb{E} [3 \cdot E_x(t+1) + n_2(t+1)]^2]$$

$$= 9 \cdot \mathbb{E} [E_x(t+1)^2] + \mathbb{E} [n_2^2(t+1)] + 6 \cdot \mathbb{E} [E_x(t+1) \cdot n_2(t+1)]$$

$$= 9 \cdot P(t) + 1 + 0$$

SEE NEXT PAGES FOR
THIS DETAILED PROOF

⇓

$$\text{Var} [E_y(t+1)] = 9 \cdot \bar{P} + 1 = 10$$

~ THIS IS CONSISTENT WITH
THIS RESULT WE GOT FROM
THE K.W. THEORY ($\text{Var} [E_y] = X^e$)

$$\begin{aligned} \mathbb{E} [e_x(t+1) \cdot n_2(t+1)] &= \mathbb{E} [(x(t+1) - \hat{x}(t+1|t)) \cdot n_2(t+1)] \\ &= \mathbb{E} [x(t+1) \cdot n_2(t+1)] + \mathbb{E} [\hat{x}(t+1|t) \cdot n_2(t+1)] \end{aligned}$$

$$\mathbb{E} [x(t+1) \cdot n_2(t+1)] = \mathbb{E} [(2/5 x(t) + n_1(t)) \cdot n_2(t+1)] = 0$$

↑
SYSTEM STATE EQUATION

$$\mathbb{E} [\hat{x}(t+1|t) \cdot n_2(t+1)] = \mathbb{E} [(1/25 \hat{x}(t|t-1) + 3/25 y(t)) \cdot n_2(t+1)] = 0$$

↑
PREDICTOR
STATE EQUATION

↑
 $y(t) = 3 \cdot x(t) + n_2(t)$